

Hand Gesture Mouse Control for Lean-Back Computing

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AIML339 Final Project Report

Abstract—We ask whether a decision-and-verification layer around an existing palm detector can drive a screen cursor by hand at 2 to 4 m in dim artificial light, where MediaPipe Hands degrades. We adopt a third-party PyTorch port of BlazePalm, train a 60,929-parameter crop verifier to replace a hand-tuned size gate, add a classical motion gate, and fine-tune 2.49% of the detector’s parameters on about 200 dark frames. Scored in one session against identical ground truth, the shipped system reaches F1 50.3% against MediaPipe’s 81.6%, with no false locks on empty-room footage for any configuration. Paired tests support the gap (exact McNemar, Holm-adjusted $p < 10^{-100}$; block-bootstrap 95% interval for the F1 difference $+0.23$ to $+0.40$). A dark-specialised variant raises VeryLowLight F1 from 10.4% to 31.5% but collapses on close static hands (93.8% to 9.4%). A staged analysis shows the palm detector loses 64.3% of target-condition hand frames, which is 86% of all frames lost once the verifier is fixed. Negative results include CLAHE and gamma correction in front of MediaPipe (F1 80.4 and 79.8 against 81.6), a logistic baseline worse than no filtering, and a motion-blob path with a 4.8-point ceiling. Verifier held-out AUC is stable across 3 to 10 seeds (SD at most 0.008); the validation-selected threshold is not.

Index Terms—hand tracking, gesture interaction, MediaPipe, BlazePalm, learned verification, low-light detection

I. INTRODUCTION

The target is a gesture mouse for lean-back computing: a webcam tracks a hand 2 to 4 m from the screen, in dim artificial light, and the hand drives the cursor. Three measured properties define the problem. First, range: the palm box is about 0.085 of the frame width. Second, lighting: our target-condition recording, VeryLowLight, has median luma 43.5 against 29 to 146 for the other clips, contrast (standard deviation) near 17 against 36 to 53, and a colour cast R/B of 1.79 against 1.05 to 1.21. MediaPipe detects a hand in only 60.8% of its frames in tracking mode and 44.7% in static mode on this clip. Third, latency tolerance is asymmetric: slow, deliberate movement is fine, but a cursor that jumps to the wrong place is not.

Research question. Does a decision-and-verification layer around a MediaPipe-class palm detector beat MediaPipe at range and in low light, and where does the remaining error come from?

Change from the proposal. The proposal was to modify MediaPipe’s tracking architecture and add an open/closed hand-state output. We abandoned that for a PyTorch port of BlazePalm, because none of the components below can be inserted into a compiled MediaPipe graph. The open/closed

output was dropped when the landmark model’s presence score fell to a median of 0.001 on VeryLowLight at locations MediaPipe itself labels as a hand (1.000 in daylight).

Contribution. This is not a new hand-tracking architecture. It is (i) a staged attribution of where a detector-based pipeline fails at range and in low light, (ii) a set of small learned and classical components, each measured on its own, and (iii) several documented failures.

II. BACKGROUND AND RELATED WORK

MediaPipe Hands and BlazePalm. MediaPipe Hands [1] uses a palm detector to propose a box, then a landmark model regresses 21 keypoints in a rotation-normalised crop. The detector is a single-shot regressor in the BlazeFace family [3], with 2944 unit-size anchors and six 1×1 heads, released in the MediaPipe graph [2]. We use the PyTorch reimplement of Satija [4], which loads Google’s released weights and is not retrained. Because its anchors are all unit-size, MediaPipe’s scale-aware anchor matching does not carry over, which matters for fine-tuning.

Data. HaGRID [5] and HaGRIDv2 [6] contain hands at 0.5 to 4 m under varied light, matching our range but with no true near-dark footage (mirror provenance: Section VI).

Baseline choice. MediaPipe is both the method we try to improve on and, through its landmark presence head, our label arbiter. Its detection output also defines the ground truth, so every result here measures agreement with MediaPipe, not with a human annotator. We also test the cheap rival a low-light report is expected to rule out: contrast enhancement in front of unmodified MediaPipe, namely CLAHE [7] on the L channel of LAB (clip limit 2.0, 8×8 tiles) and luminance gamma correction at $\gamma = 0.5$. Both use textbook parameters fixed in advance. Neither helps (Section V).

III. METHODOLOGY AND IMPLEMENTATION

Provenance. Reused: the BlazePalm architecture and released weights (1,763,358 parameters), MediaPipe as baseline, ground truth and label arbiter, and PyTorch and OpenCV. Modified: the six 1×1 prediction heads (43,966 parameters, 2.49%), fine-tuned for the dark variant only, with the original weights kept byte-identical. Extended: the acquisition motion gate and the relaxed size gate it makes safe, the motion-blob proposals, and two consensus rules. Built here: the CNN verifier and its data pipeline, the logistic baseline, the state

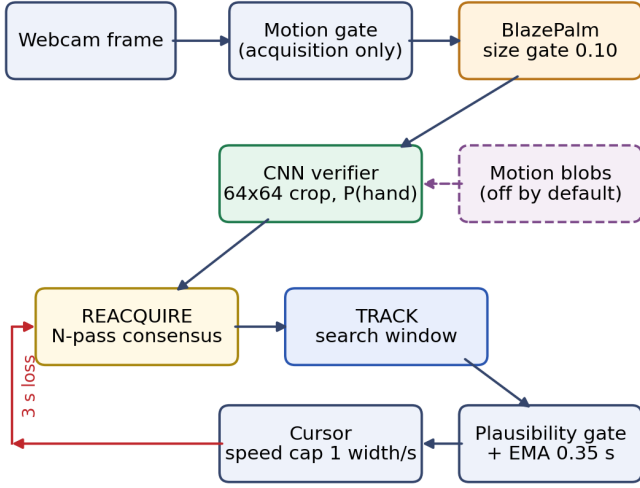


Fig. 1. Pipeline. The verifier confirms every candidate from the detector or the blob path before it can move the cursor. The motion gate acts during acquisition only.

machine with plausibility gate, smoothing and cursor control, and the evaluation harness with arbiter-based label cleaning.

A. Palm detector and fine-tune

The detector takes a 256×256 letterboxed frame scaled to $[-1, 1]$ and decodes boxes from 2944 anchors. The port discards scores below 0.7 before weighted NMS. For the dark variant only, we freeze everything except the six 1×1 heads and fine-tune on about 200 VeryLowLight frames (Adam, lr 3×10^{-4} , batch 4, 20 epochs, seed 0). The loss is $\mathcal{L}_{cls} + 0.5 \mathcal{L}_{reg}$, with positive weight 5.0 and a masked smooth-L1 over the box and two of the seven keypoints. Labelling every anchor inside the ground-truth box (about 96 of 2944) positive destroyed the score head, which fired on 337 anchors instead of 13, so the shipped version matches only the 6 anchors nearest the box centre, and a guard refuses to save a model firing on more than 100 anchors.

B. Crop verifier

A small CNN (four stride-2 Conv–BatchNorm–ReLU blocks, $3 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 64$ channels, global average pooling, one linear output; 60,929 parameters) takes the 64×64 RGB rotation-normalised crop the pipeline already computes and outputs $P(\text{hand})$. It replaces a box-size gate that was the main source of false locks at range. Training uses BCE, Adam, lr 2×10^{-3} , batch 64 on a class-balanced subsample: 8 epochs for the shipped models, and 15 epochs with the checkpoint chosen by validation loss in the seed campaign (Section V-E). Four variants were trained on 11k, 130k (arbiter-filtered), 32k (VeryLowLight-specific) and 84k (dark-augmented) crops. Photometric and distance augmentation is applied identically to both classes so brightness and sharpness cannot serve as shortcuts.

Positives for the large sets were mined as the detector’s top detection on a gesture-labelled image, which confuses *the*

image contains a hand with this detection is the hand. An independent arbiter (the landmark model’s presence head, used offline) rejected 41.6% of 30,000 mined crops, reproduced at 41.9% on 31,006 further detections.

C. Motion gate, blobs and cursor logic

Every remaining false lock sat on a static object, while even a held-still hand has micro-tremor. The motion gate acts during acquisition only: over a 5-frame, 320-px-wide luma ring, it takes the fraction of pixels whose bias- and gain-compensated value changes by more than $12/255$. Mean absolute difference and temporal standard deviation failed to separate a held-still hand (score 0.036 to 0.083) from the noisiest static clutter tile. The changed-pixel fraction separates them with at least a $3.6\times$ margin and is immune to a simulated $\pm 15\%$ auto-exposure ramp. This allowed the acquisition size gate to drop from 0.15 to 0.10. Connected components of the same signal propose motion blobs, confirmed by the same verifier.

The state machine alternates between REACQUIRE (full-frame detection plus motion gate) and TRACK (a search window around the last position, giving the detector more effective pixels on a far hand). A plausibility gate rejects jumps above 3 frame-widths per second, an EMA ($\tau = 0.35$ s) smooths positions, and the cursor is capped at 1 screen-width per second. The dark variant adds 3-pass spatial consensus on acquisition and a TRACK hold: a candidate more than 0.06 frame-widths from the committed position waits for 3 agreeing candidates or 6 frames.

IV. EXPERIMENTAL SETTINGS

Baselines. MediaPipe Hands (0.10.14, tracking mode) is the incumbent and the method we modify. A logistic regression on 8 interpretable crop statistics (9 parameters) is the simple learned baseline for the verifier. The box-size gate is the “before” state for the verifier ablation, and unmodified BlazePalm weights are the “before” state for the fine-tune.

Data and splits. Eight clips were recorded by the author: CloseLightMoving, CloseLightStill, CloseDarkStill, FarLight, FarDark, VeryLowLight (651 frames, the target condition), and two empty-room clips. Truth is MediaPipe’s static-mode output, lag-free, with gaps filled within ± 5 frames, giving 1,319 truth frames in the six hand clips (1,553 frames). Precision, recall and F1 are per frame; accepted frames are those where the system reports a position. Verifier training data come from HaGRIDv2 (11k to 130k crops, with HaGRID-level validation and test splits held out) plus 100 crops from the author’s room.

VeryLowLight is split into 50-frame blocks dealt to three roles by a seeded shuffle (split seed 0): 250 training, 151 validation and 250 test frames. Model and threshold selection use training and validation data only, and the test blocks are scored once. Three limits remain. (1) Brightness drifts across the recording, so blocks are not exchangeable: dark v3 scores F1 14.6 on the validation blocks and 43.5 on the test blocks, and we report both. (2) The end-to-end F1 in Fig. 2 covers the whole clip, including the fine-tuning blocks, so it is not a

clean test. The test blocks alone give MediaPipe 81.2, default 9.0, big verifier 10.8 and dark v3 43.5, but the shipped dark v3 was fine-tuned on an earlier two-way split (203 frames, 88 held out) whose training blocks may overlap these test blocks. Only the seed-campaign detector results (Section V-D; 129 training frames) are clean held-out evidence for the fine-tune. The other five hand clips were not used for fine-tuning, though several operating parameters were chosen on them. (3) All dark data is one person, room, lamp, camera and session.

Parameters. Settings were varied one at a time where swept: the acquisition size gate (0.15, 0.10, 0.08, 0.06; chosen 0.10), the motion-gate floor (0.01, between a 0.036 weakest still hand and 0.000 for a static room), and the verifier thresholds (0.20 to 0.50, chosen 0.30 by clutter rejection at fixed recall; 0.61 for the 130k model, matched to the default’s own-room clutter cut of 91.3%). Detector positive weight 20.0 collapsed the head, so 5.0 is used. Learning rates, batch sizes and epochs follow small-model defaults and were not swept. The consensus rules were read off measured frame-to-frame displacements; the plausibility limit, EMA τ and cursor cap are design choices.

Seeds, determinism and hardware. All randomised components use recorded seeds: split seed 0, training seeds 0 to 9 for the verifier variants and detector fine-tune, and bootstrap seed 12,345. No seed comes from the clock. Section V-E reports how many seeds each component completed.

An earlier harness gave F1 varying by about ± 1.3 points between runs of the same fixed model, which we had called a noise floor. It was not model variance: the plausibility gate and tracking-loss timer are time based, and the replay harness fed them wall-clock Δt . Timing now comes from the frame index and the clip’s frame rate. Three repeats of one configuration give byte-identical output on all 16 clip and pipeline rows (only milliseconds per frame differ), so the evaluation noise floor is exactly zero and all seven systems are scored in one session against one truth. Earlier-round component figures (label errors, colour cast, ablations, timings) predate the timing fix and are in the repository design notes. Hardware: AMD Ryzen 5 7500F (6 cores), no GPU, Windows 10, Python 3.10.20, PyTorch 2.13.0+CPU, OpenCV 5.0.0.

V. RESULTS AND ANALYSIS

A. Comparison with MediaPipe

Fig. 2 and Table I give the main result. MediaPipe leads in aggregate (F1 81.6% against 45.4 to 52.5%), mostly through recall (75.7% against 32.7 to 40.0%). At close range the gap narrows to 6 to 14 F1 points for the default and big configurations, and all seven systems score 0.0% false locks on both empty-room clips. The 130k-crop verifier at 0.61 more than doubles the default’s VeryLowLight F1 (10.4 to 21.5), though its aggregate advantage is not separable from noise (Section V-F).

Contrast preprocessing does not help. CLAHE in front of unmodified MediaPipe reaches F1 80.4 and gamma 0.5 reaches 79.8, both below raw 81.6. CLAHE is correct on

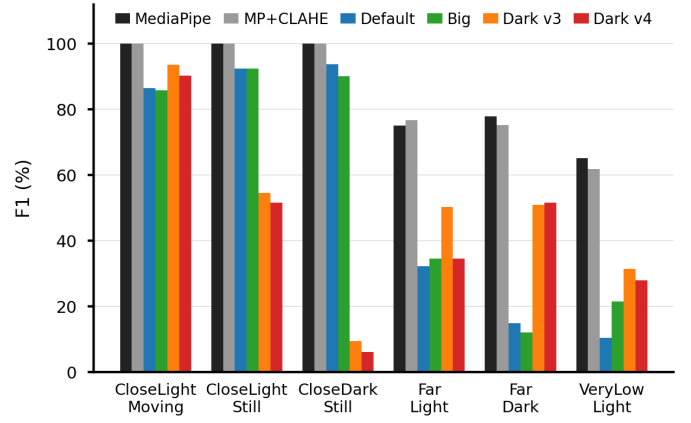


Fig. 2. F1 (%) per clip, one session, identical ground truth. +C = MediaPipe behind CLAHE; Big = 130k-crop verifier at 0.61; Dark v3 = fine-tuned detector plus acquisition consensus; v4 adds the TRACK hold. VeryLowLight includes the fine-tuning blocks.

TABLE I
AGGREGATE RESULTS, SAME SYSTEMS AS FIG. 2. FL = FALSE-LOCK RATE (%) ON THE TWO EMPTY-ROOM CLIPS.

	MP	+C	Default	Big	Dark v3	Dark v4
Aggregate F1	81.6	80.4	50.3	52.5	48.8	45.4
Precision	88.3	88.9	79.2	77.0	62.6	74.2
Recall	75.7	73.4	36.8	39.9	40.0	32.7
FL Light/Dark	0/0	0/0	0/0	0/0	0/0	0/0

29 frames where raw MediaPipe is wrong and wrong on 55 where raw is right (exact McNemar $p_{\text{Holm}} = 0.018$). Both lose on VeryLowLight (65.1 raw, 61.8 CLAHE, 62.2 gamma). The target-condition failure is not a contrast deficit, consistent with the attribution below.

B. Where the target-condition loss occurs

Of the 291 VeryLowLight frames where lag-free MediaPipe sees a hand (a stricter count than the 486 truth frames used for F1), the detector fails to produce a hand-containing box on 187 (64.3%): 77 with no detection and 110 off the hand. The size and motion gates remove 18 more. With the default verifier, 70 of the 86 crops that reach it are rejected (81%) and only 16 frames (5.5%) survive. With the 130k verifier, 13 are rejected (15%) and 73 (25.1%) survive. The detector therefore accounts for 68% of all lost frames under the default verifier and 86% (187 of 218) once the verifier is fixed.

This is not a confidence problem. Any detection appears on 59.8% of frames at every score threshold from 0.3 to 0.7, with median best score 0.76: the detector fires confidently, just often not on the hand. On clean, well-lit HaGRID images its top detection is off the hand in about 42% of cases, so the failure is not specific to our footage.

C. Ablations

On the 7-clip integrated evaluation (five hand clips plus two empty-room clips; hand frames kept, clutter locks), the box-size gate alone keeps 499 hand frames with 156 clutter

locks, the logistic baseline 379 with 190, and the CNN verifier 490 with 143 at threshold 0.20, 486 with 32 at 0.30 (chosen) and 461 with 9 at 0.50. These operating points come from the earlier harness, before the timing fix; re-running only the chosen configuration gives 451 and 33.

Verifier. At 0.30 the CNN keeps 97.4% of true hand frames while cutting clutter locks by 79% (same earlier-harness run). The logistic baseline is worse than no filtering (120 hand frames lost, 34 clutter locks added), with held-out AUC 0.788 against 0.965 for the CNN on the earlier split (ten-seed mean on the HaGRID test crops: 0.975 ± 0.005). The 130k verifier raised hand recall on VeryLowLight crops from 4.9% to 58.5% at matched clutter rejection, at lower portability (HaGRID clutter cut 91.5% to 70.9%, AUC 0.965 to 0.947 on the earlier split; 0.975 to 0.951 across seeds).

Motion gate. Alone it removes 29 of 32 clutter locks but costs 48 hand frames, because it correctly refuses to bootstrap far-range acquisitions from clutter. Relaxing the size gate to 0.10, safe only because the motion gate now does the filtering, recovers 31 frames (469 kept) and cuts clutter locks 53% against the ungated verifier (15 against 32).

Motion blobs. Where the detector fails outright, blobs propose a hand in 21.4% of cases and the blob verifier confirms only 7.5% of those, a ceiling of about 14 frames (4.8 points). Without a blob verifier they raise VeryLowLight clutter locks from 98 to 405 and lock onto a synthetic moving non-hand object every time, so blobs are off by default. In the dark variant they stay on, because they enable verifier-gated acquisition: turning them off raises the no-hand false-lock rate from 1.0% to 12.3%.

D. Detector fine-tune and consensus

Fine-tuning gives the largest single gain, and it is a redistribution. With the fine-tuned detector plus acquisition consensus (Dark v3), VeryLowLight F1 rises 10.4 to 31.5 (28.0 with the TRACK hold), FarDark 14.9 to 50.9 and FarLight 32.3 to 50.2, but CloseDarkStill falls 93.8 to 9.4 and CloseLightStill 92.5 to 54.6. FarDark and FarLight are the better evidence for generalisation, since those clips were not used for fine-tuning. On the 106 test-block frames that contain a hand, the unmodified detector puts a box on the hand in 30.2% and five fine-tune seeds in 45.5% (SD 2.0). On the validation blocks the order reverses (80.4% against 65.7%), more evidence that blocks are not exchangeable and that validation loss is a weak checkpoint criterion here. An earlier build of the dark variant measured 2.2% false locks on NohandLight, which does not reproduce under fixed timing; we report it because the number changed.

The TRACK hold (Dark v3 to v4) cuts off-hand cursor placements from 316 to 150, wrong-location excursions from 79 to 30 (62% fewer) and persistent wrong locks from 7 to 1, at a cost of 7.3 points of recall and 3.4 of F1 (48.8 to 45.4). F1 counts frames with a reported position, so holding the cursor lowers it by construction, and a cursor that is briefly still is a different defect from one that jumps: F1 alone undervalues this change, and an excursion metric alone would hide the recall

TABLE II
VERIFIER VARIANTS RETRAINED FROM SCRATCH WITH THE SHARED SEED LIST, MEAN $\pm$ SD OVER COMPLETED SEEDS. HaGRID AUC IS ON HaGRID TEST CROPS; TARGET AUC IS ON THE VERYLOWLIGHT TEST BLOCKS (BLANK WHERE THE POOL LACKS SUCH CROPS). THR. IS THE OPERATING POINT SELECTED ON THE 151 VALIDATION FRAMES.

Variant	Seeds	HaGRID AUC	Target AUC	Thr.	Epoch	Train s
11k mined	10	0.975 ± 0.005	—	0.08 ± 0.06	7.8 ± 4.0	25
130k arbitered	3	0.951 ± 0.008	0.64 ± 0.07	0.10 ± 0.07	2.0 ± 1.7	576
Target-cond.	9	0.972 ± 0.006	0.71 ± 0.08	0.23 ± 0.16	7.1 ± 4.3	22
84k dark-aug.	0		not run			

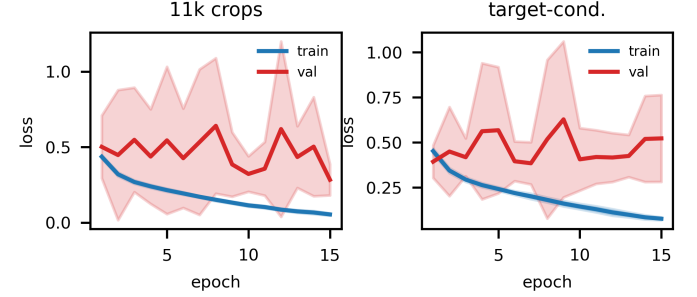


Fig. 3. Verifier training and validation loss, mean and range over seeds, for the 11k and target-condition variants. Training loss falls smoothly; validation loss is noisy because it is computed on 151 frames.

cost. Acquisition consensus on its own is neutral end to end, and on CloseDarkStill it fails structurally: false detections are stable and true ones transient, so agreement over time discards the truth. That clip needs a crop-level discriminator, not more timing logic.

E. Training runs across seeds

Ten seeds were planned for each verifier variant and the detector fine-tune, and the campaign was stopped early: ten seeds for the 11k variant, nine for the target-condition variant, three for the 130k variant, none for the 84k variant, and five for the detector. The per-seed end-to-end pipeline evaluation was not run, so Table II reports what the seeds vary: held-out AUC, the epoch selected by validation loss, and the threshold chosen on the validation blocks. (This validation-F1 threshold differs from the clutter-rejection thresholds of Section IV.)

The weights are stable: held-out AUC varies by 0.005 to 0.008 across seeds, three or four times smaller than the 0.02 gap between the 130k variant and the two small ones, so that ranking is not seed noise. The operating threshold is not stable: it ranges from 0.05 to 0.50 (SD 0.06 to 0.16 within a variant), because 151 validation frames cannot pin it down, and the noisy validation loss in Fig. 3 means the selected epoch is weakly determined too. Target AUCs of 0.64 and 0.71 are low, with a gap within one SD.

F. Domain dependence and statistical position

The dark verifier keeps 61.3% of held-out hand crops in the recorded colour cast but only 4.0% with the cast removed (38.7% warmer, 52.4% cooler). It has partly learned this lamp’s colour, which is also a bias risk.

TABLE III
PAIRED TESTS OVER THE SIX HAND CLIPS (1,553 FRAMES). A/B COUNTS FRAMES WHERE A DECIDED CORRECTLY AND B DID NOT, AND THE REVERSE. p IS EXACT MCNEMAR, HOLM-ADJUSTED OVER TEN COMPARISONS. $\Delta F1$ HAS A 95% BLOCK-BOOTSTRAP INTERVAL. A DAGGER MARKS A FRAME-LEVEL DIFFERENCE THAT IS SIGNIFICANT WHILE THE F1 INTERVAL SPANS ZERO.

A	B	A/B	p (Holm)	$\Delta F1$	95% CI	Sig.
MP raw	default	553/38	3e-117	+0.31	+0.226, +0.403	yes
MP raw	big	538/60	3e-96	+0.29	+0.195, +0.384	yes
MP raw	dark v4	638/57	3e-124	+0.36	+0.281, +0.445	yes
MP raw	MP CLAHE	55/29	0.018	+0.01	-0.002, +0.025	yes†
MP raw	MP gamma	66/30	2e-03	+0.02	+0.002, +0.033	yes
default	big	54/91	0.011	-0.02	-0.066, +0.014	yes†
big	dark v3	262/216	0.079	+0.04	-0.036, +0.103	no
dark v3	dark v4	112/55	7e-05	+0.03	+0.001, +0.067	yes

The per-frame decisions of all seven systems were kept, so they are paired binary outcomes on the same frames. We use the exact McNemar test on discordant pairs, and a block bootstrap over 50-frame blocks within each clip (2,000 resamples, seed 12,345) for a 95% interval on F1 differences, since F1 is not a mean of independent frames. The ten comparisons form one family, Holm-Bonferroni adjusted, $\alpha = 0.05$. Table III shows eight; the other two are MP against dark v3 (608/84, $p_{\text{Holm}} = 5 \times 10^{-98}$) and CLAHE against gamma (43/33, $p_{\text{Holm}} = 0.302$, not significant).

The central comparison is supported without qualification: each of our four configurations differs from MediaPipe at $p_{\text{Holm}} < 10^{-95}$, with an interval nowhere near zero. Big against dark v3 is not significant, so Big’s 3.7-point aggregate lead over dark v3 is not claimed. In the daggered rows, CLAHE loses more frames than it wins (55 against 29) without a clear F1 effect, and the 130k verifier decides correctly on 91 frames where the default does not against 54 the other way, a frame-level gain F1 alone would hide. On one pool of 4,071 held-out crops, AUC is 0.860 unfiltered and 0.959 arbitered.

G. Computational cost

BlazePalm has 1,763,358 parameters (7.1 MB); the verifier adds 60,929 (255 KB) and the fine-tune adds none. The pipeline runs at 19 to 22 ms per frame on CPU with blobs off and 21 to 40 ms with blobs on, against 0.2 to 0.3 ms for MediaPipe’s cached detection step: two orders of magnitude slower, but about 50 fps, enough for slow gestures. Verifier training took 22 to 576 s per variant and the detector fine-tune 222 s.

VI. ETHICAL CONSIDERATIONS

Licensing and provenance. HaGRID is released under CC BY-SA 4.0 [5]. We obtained HaGRIDv2 from two third-party mirrors (zhouxzh/portable-hagridv2-mediapipe-hand and testdummyvt/hagridv2_512px) rather than the official release, so completeness and licence-file preservation are not guaranteed. Whether a model trained on CC BY-SA images is “adapted material” is unsettled; we state the licence without a legal conclusion. No HaGRID images appear here.

Privacy. All eight clips show only the author’s hand and room, recorded with consent; HaGRID’s subjects consented to that dataset’s release. The application keeps frames in memory and makes no network calls.

Bias and representation. Dark data comes from one person, room, lamp and camera, so no cross-user or cross-environment generalisation is shown. Label cleaning uses MediaPipe’s landmark head as arbiter, so any blind spot of that model (skin tone, hand size, unusual posture) is inherited by our data and is not characterised.

Safety, misuse and cost. Misuse potential is low: the output is local cursor coordinates. The realistic risk is presenting this as an accessibility aid. The default system reports a position on only 39.5% of hand-clip frames (614 of 1,553), has wrong-location excursions, and was trained on one user, so it is a classroom prototype, not validated for people who depend on it. The only user-facing stop is the pyautogui corner fail-safe, and the motion gate assumes a stationary camera. Development used about 10 GB of downloads and 1.4 million generated crops; energy was not measured.

VII. CONCLUSIONS AND FUTURE WORK

Answer. Partly, and only in specific conditions. MediaPipe wins in aggregate (F1 81.6% against 45.4 to 52.5%), and paired tests confirm the gap. At close range the gap is 6 to 14 points for the default and big configurations, with zero false locks for every configuration; the dark configurations are about 45 points behind on CloseLightStill. In the target regime the dark variant raises VeryLowLight F1 from 10.4 to 31.5 over the default (and from 30.2% to 45.5% hand-containing boxes on held-out blocks over five seeds), still trailing MediaPipe’s 65.1 by 34 points on the whole clip. The gain is a redistribution, with a close-static collapse from 93.8 to 9.4. Two plausible interventions failed: contrast enhancement in front of MediaPipe, and motion proposals aimed at the detector’s misses. The binding constraint is the palm detector, which loses 64.3% of target hand frames before any added component acts, and our one narrow intervention there gave the largest gain.

Limitations. One dark session, one person; 3 to 10 verifier seeds and 5 detector seeds, with end-to-end F1 from single runs; MediaPipe-derived ground truth, so every number measures agreement with MediaPipe; validation blocks unrepresentative of test blocks, so every threshold is one draw from a wide interval; no click vocabulary. The seed campaign was stopped by time, not compute, which is the weakest part of this evaluation.

Future work. (1) Finish the seed campaign with per-seed end-to-end evaluation. (2) Use more validation frames for thresholds. (3) Hand-label a subset of VeryLowLight to break the circular ground truth. (4) Fine-tune the detector on broader artificial-light data and record a second dark session in a different room. (5) Add a crop-level discriminator for stable false positives.

STATEMENT

This report and the project it describes are my own work. Code: <https://github.com/Filkspline/AI339ProjectRepo>. Tools: Python, PyTorch [8], OpenCV [9], MediaPipe [2], pyautogui, PyInstaller, $\LaTeX$.

Generative AI: DeepSeek’s coding harness was used to set up the project file structure and help with implementation. I am responsible for all content. External resources: BlazePalm [4] and its released weights [2], MediaPipe Hands [1], and HaGRIDv2 [6] via the two mirrors named in Section VI.

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